

Programming for Robotics: Project Report

Project Title: SMART HAZARD
DETECTION SYSTEM

GitHub link: <https://github.com/MahakaliAyush/smart-hazard-detection-system>

Wokwi Simulation Link: <https://wokwi.com/projects/461326802980797441>

Video Link: <https://youtu.be/MOTOaN9Hq4k?si=bCaUvqBzGGshPnMd>

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Specialisation Option Chosen:

Embedded System (Wokwi)

1. Project Overview

The Smart Hazard Detection System is an embedded intelligent monitoring system designed to detect hazardous environmental conditions in real time. The system is intended for high-risk environments such as industrial facilities, storage areas, and fire-prone zones where early hazard detection is important for safety and prevention.

The project was implemented using the ESP32 microcontroller within the Wokwi simulation environment. The system combines multiple sensors and intelligent decision-making using a Finite State Machine (FSM) to classify environmental conditions into different operational states.

The system uses a DHT22 sensor to monitor temperature and humidity levels, while an HC-SR04 ultrasonic sensor detects nearby objects and environmental presence. Based on the sensor data, the system reacts using multiple outputs including RGB-style LED indicators, a buzzer alarm system, and a servo motor for intelligent scanning behaviour.

The implemented intelligent behaviour includes sensor fusion, adaptive monitoring intervals, multi-condition decision logic, and fail-safe protection mechanisms. These features allow the system to respond differently depending on the severity of environmental conditions while maintaining stable and reliable operation.

2. Technical Requirements Compliance

The project satisfies all required technical specifications outlined in the assessment brief.

Inputs:

The system uses two input sensors:

- DHT22 sensor for temperature and humidity measurement
- HC-SR04 ultrasonic sensor for distance and presence detection

Outputs:

The system implements multiple outputs:

- Green, Yellow, and Red LED indicators
- Buzzer alert system
- Servo motor scanning behaviour

Finite State Machine:

The system uses a structured Finite State Machine (FSM) to control behaviour and decision-making.

Behavioural States:

The FSM contains five operational states:

- IDLE
- SURVEY
- CAUTION
- EMERGENCY
- FAILSAFE

Multi-Condition Decision Logic:

The system uses sensor fusion and multi-condition evaluation:

- CAUTION state triggers when temperature exceeds 45°C OR humidity drops below 20%
- EMERGENCY state triggers when temperature exceeds 60°C AND humidity drops below 15%

Safety and Fail-Safe Mechanism:

The project includes a FAILSAFE state that activates when invalid sensor readings or abnormal conditions are detected. This ensures stable and safe operation under sensor failure conditions.

Structured System Architecture:

The system follows a Sense → Process → Decide → Act architecture model using modular software design and separated behavioural functions.

3. System Architecture

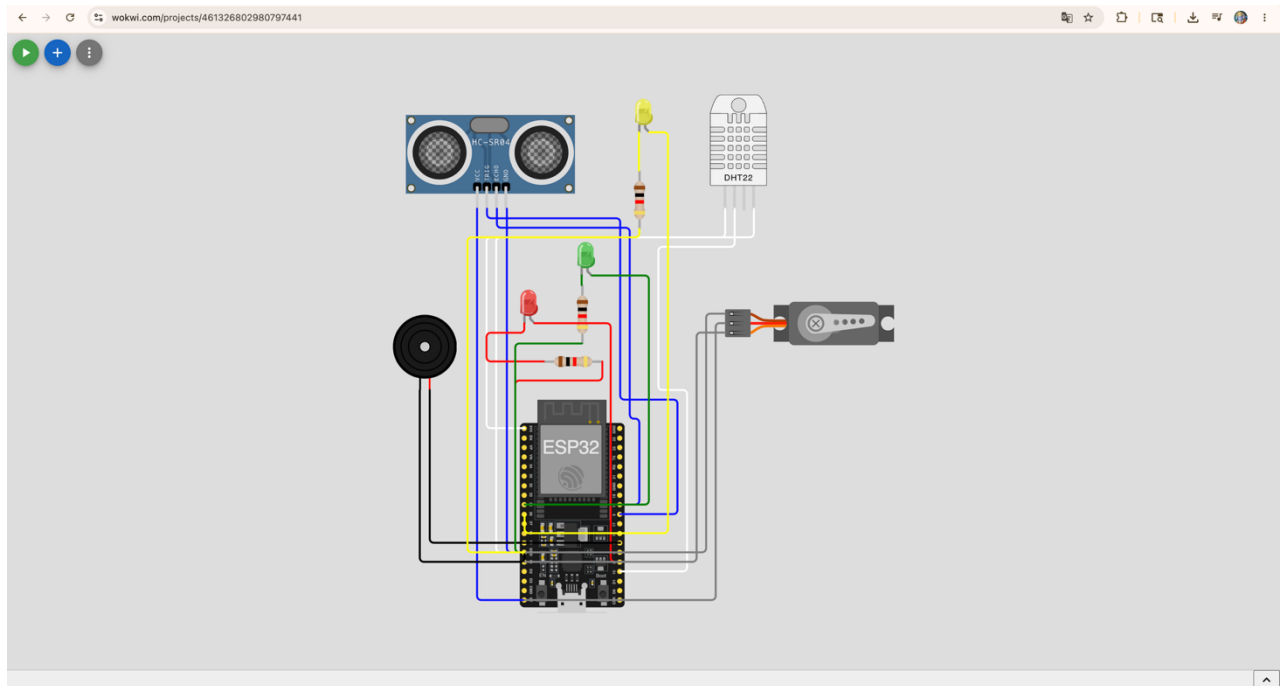


Figure 1- Full System

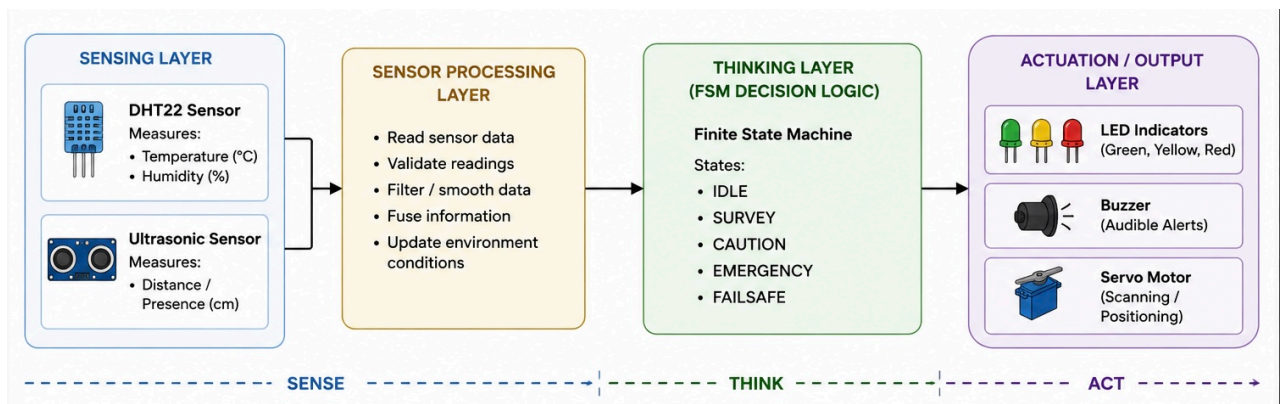


Figure 2 - Architecture Diagram

The Smart Hazard Detection System follows the Sense → Process → Decide → Act robotics architecture model.

Sense Layer:

The sensing layer gathers environmental information using:

- DHT22 sensor for temperature and humidity monitoring
- HC-SR04 ultrasonic sensor for object distance detection

Process Layer:

Sensor data is processed and validated by the ESP32 microcontroller. The system continuously evaluates environmental conditions and applies threshold-based decision logic.

Decision Layer:

The Finite State Machine (FSM) analyses the processed sensor data and determines the appropriate system state:

- IDLE
- SURVEY
- CAUTION
- EMERGENCY
- FAILSAFE

Actuation Layer:

Based on the selected state, the system controls:

- LED indicators for visual feedback
- Buzzer for audible alerts
- Servo motor for intelligent scanning movement

The information flow moves continuously from sensor acquisition to processing, decision-making, and output actuation in real time.

4. Behaviour Model

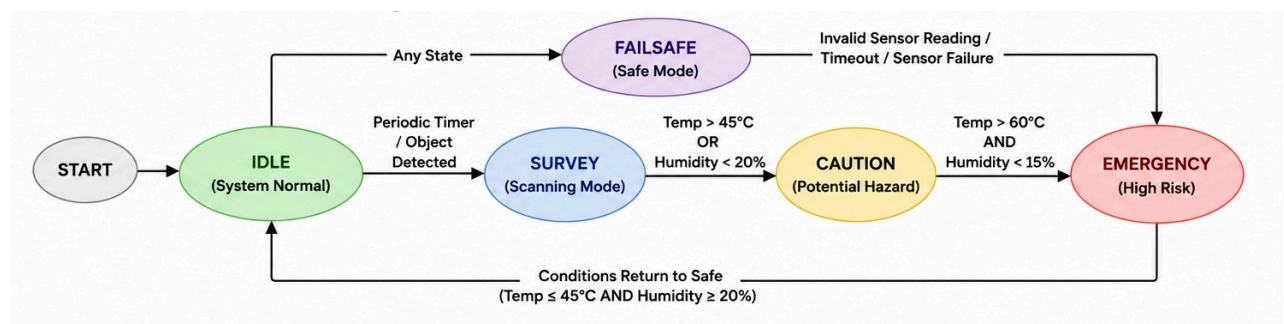


Figure 3 - FSM Diagram

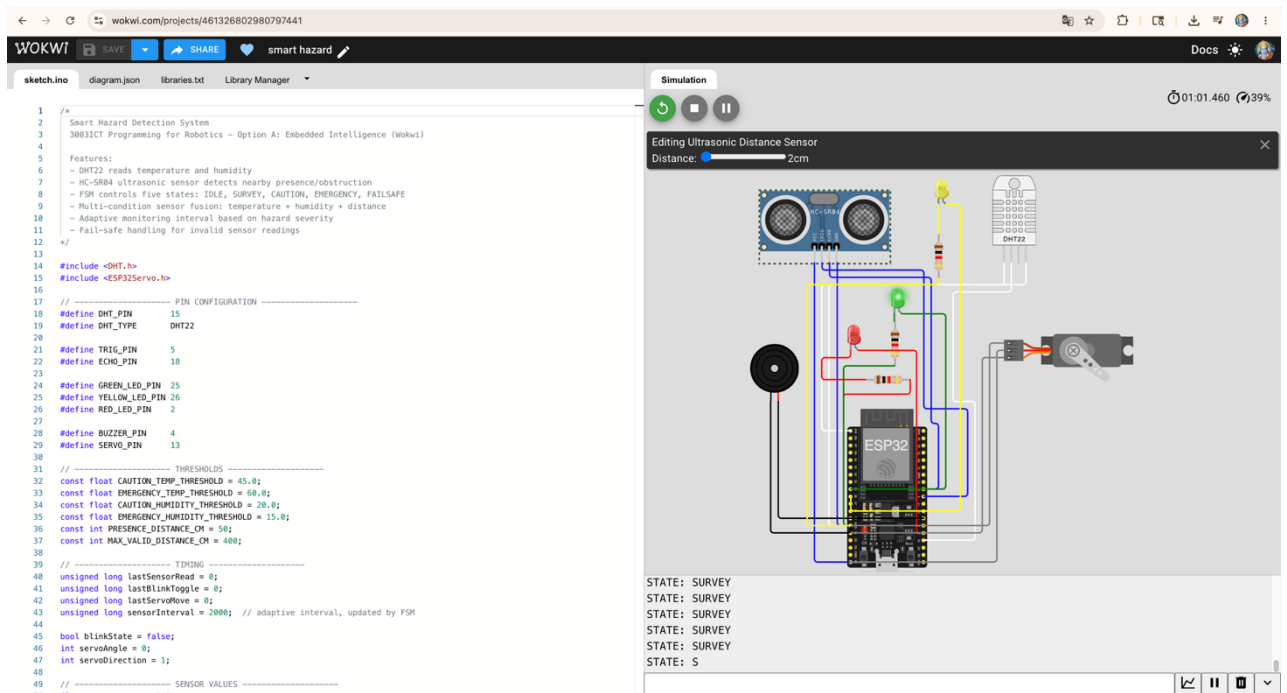


Figure 4 - Survey State

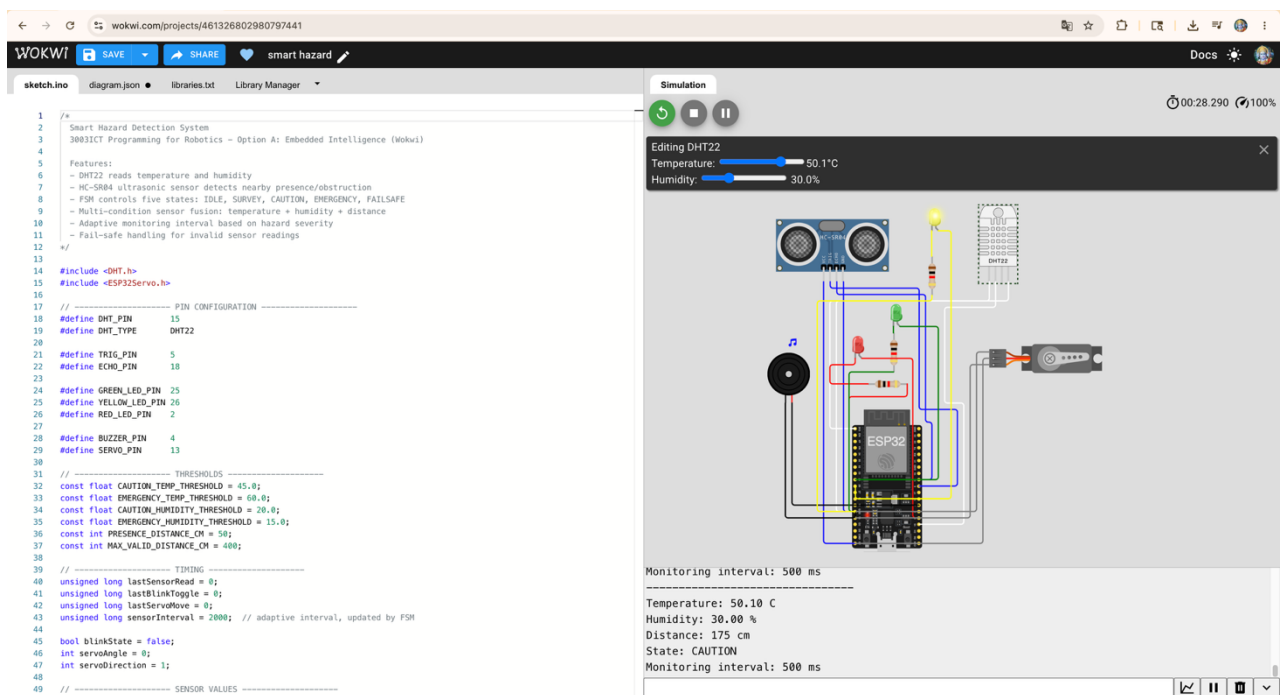


Figure 5 - Caution State

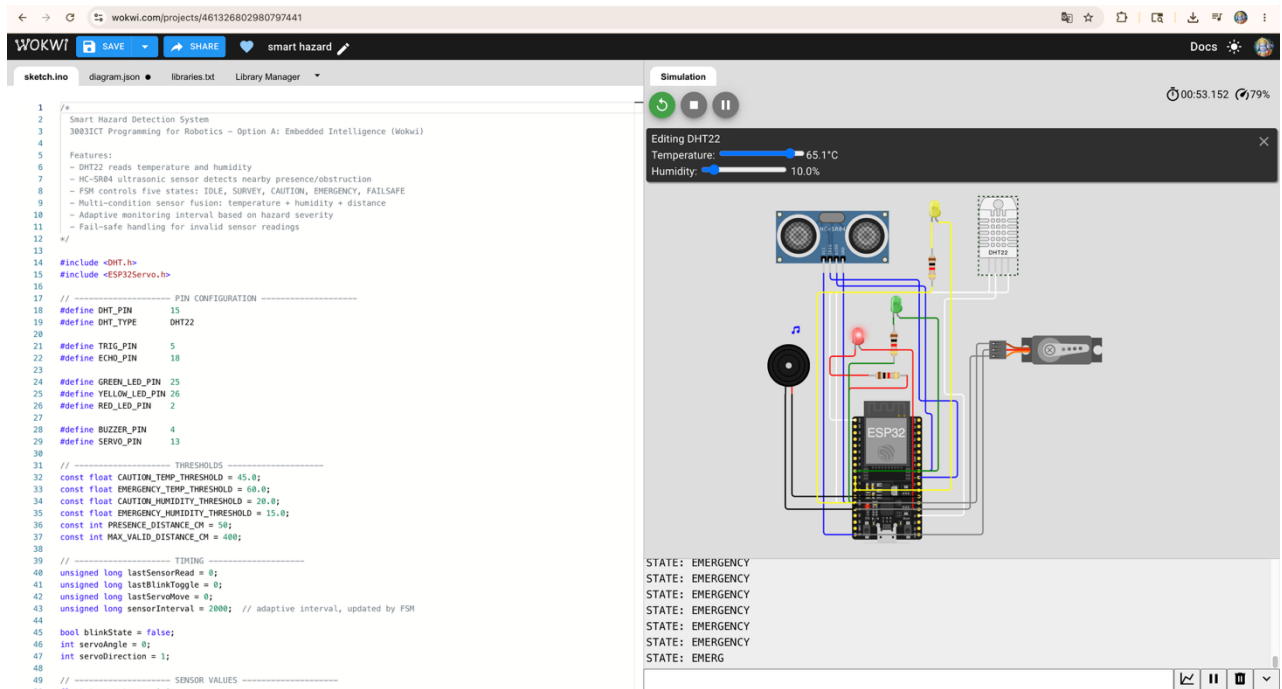


Figure 6 - Emergency State

The system behaviour is controlled using a Finite State Machine (FSM). This approach was selected because FSMs provide structured, predictable, and reliable control logic for embedded intelligent systems.

The implemented FSM contains five operational states:

IDLE:

The system enters the IDLE state when environmental conditions remain within safe thresholds. In this state, the green LED remains active, the buzzer is disabled, and the servo motor remains centred while low-frequency monitoring occurs.

SURVEY:

The SURVEY state performs active environmental scanning. The servo motor continuously sweeps between 0° and 180° to simulate intelligent monitoring behaviour while the system performs active sensing operations.

CAUTION:

The CAUTION state activates when:

- Temperature exceeds 45°C
- OR
- Humidity falls below 20%

In this state, the yellow LED blinks slowly and the buzzer generates intermittent warning tones. This behaviour indicates the presence of a potential environmental hazard.

EMERGENCY:

The EMERGENCY state activates when:

- Temperature exceeds 60°C

AND

- Humidity falls below 15%

The system responds with rapid red LED flashing, continuous buzzer alerts, and restricted servo positioning to indicate a high-risk hazardous condition.

FAILSAFE:

The FAILSAFE state activates when invalid sensor readings, timeout conditions, or abnormal sensor behaviour are detected. In this state, the system enters a restricted safe operating mode with emergency visual and auditory alerts enabled.

The FSM structure ensures clear behavioural transitions, stable operation, and easy debugging while improving overall system reliability and responsiveness.

5. Perception / Sensor Processing

The Smart Hazard Detection System uses multiple sensors and sensor fusion techniques to analyse environmental conditions in real time.

The DHT22 sensor continuously measures:

- Temperature (°C)
- Humidity (%)

The HC-SR04 ultrasonic sensor measures:

- Distance to nearby objects
- Presence detection within a predefined monitoring range

Sensor data is processed by the ESP32 microcontroller using threshold-based evaluation and multi-condition logic. The system combines temperature and humidity readings to determine hazard severity instead of relying on a single sensor value.

The CAUTION state uses OR-based logic:

- Temperature > 45°C OR Humidity < 20%

The EMERGENCY state uses AND-based logic:

- Temperature > 60°C AND Humidity < 15%

This sensor fusion approach improves system intelligence by reducing false alarms and enabling context-aware behaviour.

The ultrasonic sensor supports environmental awareness by detecting nearby objects and triggering active scanning behaviour during the SURVEY state.

The system also implements adaptive monitoring intervals. Under safe conditions, sensor readings occur at slower intervals to reduce unnecessary processing. During hazardous conditions, monitoring frequency automatically increases to provide faster system response and higher real-time accuracy.

6. Safety and Robustness

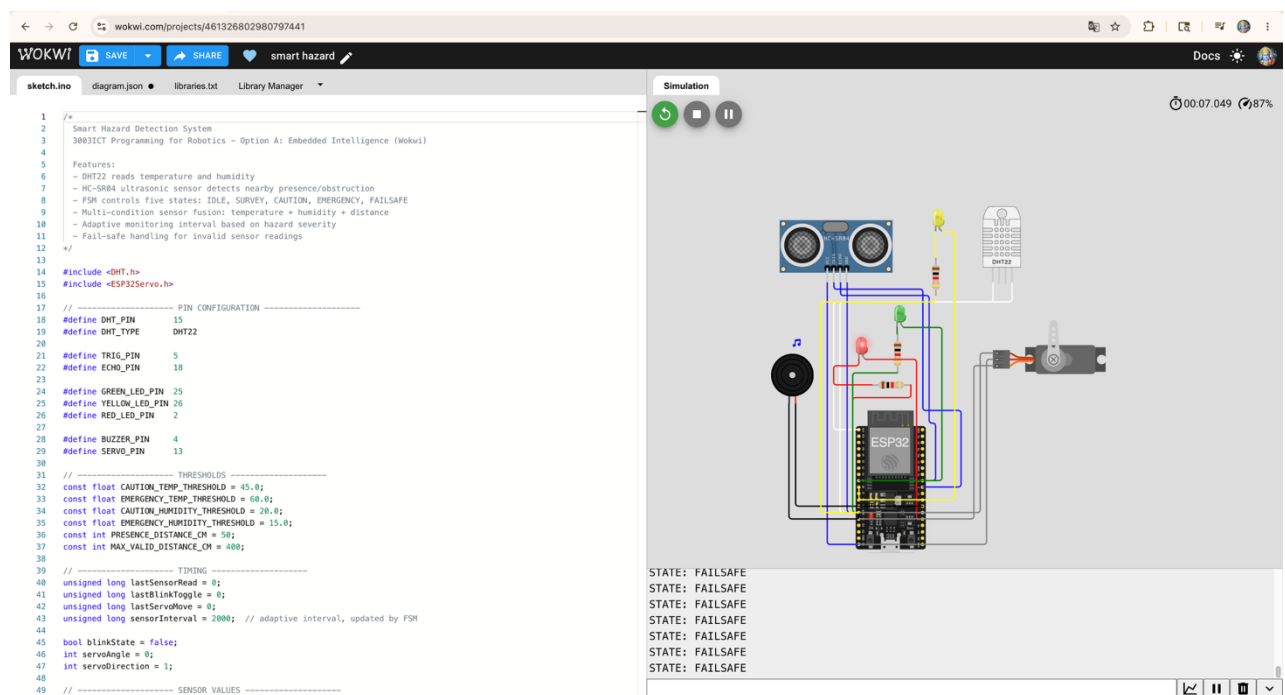


Figure 7 - Fail-safe State

The Smart Hazard Detection System includes several safety and robustness mechanisms to ensure reliable operation under different environmental conditions.

Fail-Safe Behaviour:

The system continuously validates sensor readings. If invalid values, timeout conditions, or abnormal sensor responses are detected, the FSM automatically transitions into the FAILSAFE state.

In FAILSAFE mode:

- Emergency alerts remain active
- Restricted servo positioning is enabled
- Visual warning indicators notify the user of abnormal system behaviour

Adaptive Monitoring:

The system dynamically adjusts monitoring frequency based on environmental risk levels. Higher-risk conditions increase sensor polling frequency to improve responsiveness and hazard detection speed.

Stable FSM Transitions:

The FSM structure prevents unpredictable state switching and ensures controlled behavioural transitions between safe, warning, and emergency conditions.

Visual and Audible Alerts:

Multiple output systems improve safety communication:

- LEDs provide clear visual indication of system status
- The buzzer provides immediate audible alerts during hazardous conditions

Robust Sensor Processing:

Sensor fusion and multi-condition logic reduce false-positive detections and improve overall system reliability compared to single-sensor systems.

7. Code Structure

The program was developed using a modular and structured software design approach to improve readability, maintainability, and debugging efficiency.

Main Components:

The code is divided into multiple functional sections including:

- Sensor reading functions
- FSM state update logic
- Output handling functions
- Servo scanning functions
- Utility/helper functions

Main Control Loop:

The main loop continuously:

1. Reads sensor data
2. Validates sensor values
3. Updates the FSM state
4. Executes output behaviour based on the current state

Key Functions:

- `readDistance()` – measures ultrasonic sensor distance
- `updateState()` – determines the current FSM state
- `handleStateOutputs()` – activates outputs based on state
- `handleIdleState()` – manages safe operating behaviour

- `handleSurveyState()` – controls servo scanning behaviour
- `handleCautionState()` – activates warning alerts
- `handleEmergencyState()` – activates emergency alerts
- `handleFailsafeState()` – manages fail-safe behaviour
- `sweepServo()` – controls continuous servo scanning motion

Libraries Used:

The project uses the following libraries:

- `DHT.h` – for DHT22 temperature and humidity sensing
- `ESP32Servo.h` – for servo motor control

The modular structure improves engineering quality and makes the system easier to expand and maintain.

8. Key Code Snippets

The following code snippets demonstrate key intelligent behaviours implemented within the system.

FSM State Decision Logic:

```
if (temperature > 60 && humidity < 15) {
    currentState = EMERGENCY;
}
else if (temperature > 45 || humidity < 20) {
    currentState = CAUTION;
}
```

This logic demonstrates sensor fusion and multi-condition evaluation using both AND and OR conditions.

Servo Scanning Behaviour:

```
void sweepServo(unsigned long now, int speedMs) {
    if (now - lastServoMove >= speedMs) {
        lastServoMove = now;
        servoAngle += servoDirection * 2;

        if (servoAngle >= 180) {
            servoDirection = -1;
        }
        else if (servoAngle <= 0) {
```

```
        servoDirection = 1;
    }

    scanServo.write(servoAngle);
}
}
```

This function allows the servo motor to continuously scan the environment during the SURVEY state.

Fail-Safe Logic:

```
if (isnan(temperature) || isnan(humidity)) {
    currentState = FAILSAFE;
}
```

This ensures the system enters a safe restricted mode if invalid sensor values are detected.

Adaptive Monitoring:

```
if (currentState == EMERGENCY) {
    monitoringInterval = 200;
}
else {
    monitoringInterval = 1000;
}
```

The monitoring frequency increases during hazardous conditions to improve real-time responsiveness.

9. Results / Demonstration Summary

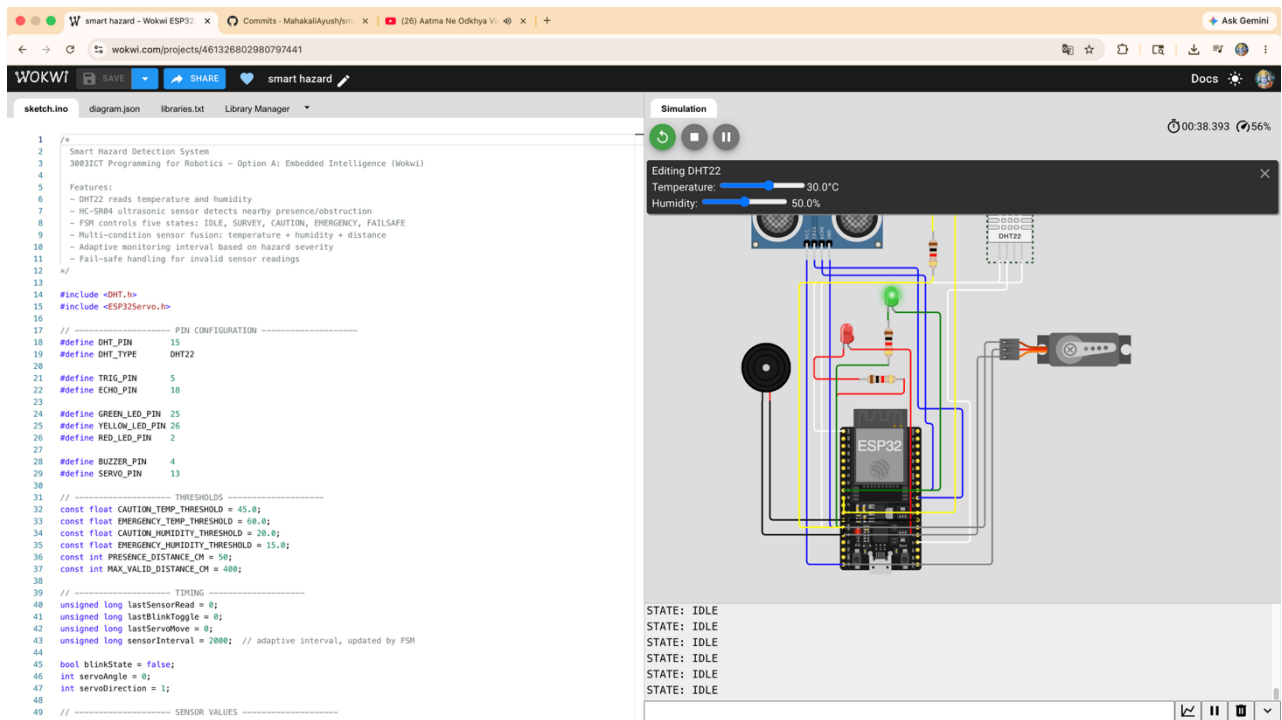


Figure 8 - Idle State

The Smart Hazard Detection System successfully demonstrated intelligent environmental monitoring and adaptive hazard response behaviour within the Wokwi simulation environment.

The system successfully:

- Monitored temperature and humidity in real time
- Detected nearby objects using the ultrasonic sensor
- Transitioned correctly between all FSM states
- Activated visual LED alerts based on hazard severity
- Generated audible buzzer warnings during caution and emergency conditions
- Performed intelligent servo scanning behaviour
- Entered FAILSAFE mode during invalid sensor conditions

The implemented sensor fusion and adaptive monitoring logic improved the overall intelligence and responsiveness of the system.

Challenges Encountered:

Several challenges were encountered during development including:

- ESP32 servo library compatibility issues
- FSM timing synchronisation
- Managing stable non-blocking behaviour using millis()
- Coordinating multiple outputs simultaneously

These issues were resolved through modular code design, testing, and iterative debugging.

10. Individual Contributions

Member	Contribution
Ayush Patel	Project design, FSM implementation, sensor integration, embedded intelligence logic, Wokwi simulation development, adaptive behaviour implementation, system debugging, architecture design, report writing, testing, and final system integration
Lachlan Roche	Wokwi circuit setup assistance, GitHub repository management, project coordination, documentation support, screenshot/media organisation, and video preparation assistance, and final submission coordination